

DEVIN RANKIN

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EDUCATION

Temple University

Bachelor of Science, Computer Science

Philadelphia, PA

Aug 2020 – May 2024

- 1x Hackathon winner
- 2x Dean's List

EXPERIENCE

Software Engineer Intern

May 2023 – Aug 2023

Coventry Life Settlements

Fort Washington, PA

- Worked within a large legacy C#/ASP.NET codebase, contributing to backend services and internal tooling.
- Rapidly prototyped and delivered AI-driven onboarding tools for IT teams, automating manual workflows and reducing setup time.
- Collaborated in an Agile team using Git, Jira, and Confluence for feature development and debugging.
- Led efforts to document a decades-old undocumented codebase, improving maintainability and onboarding for future developers.

Teaching Assistant / CS Peer Tutor

Feb 2023 – May 2024

Temple University

Philadelphia, PA

- Supported 70+ students in operating systems and low-level programming through lab instruction and one-on-one guidance.
- Led weekly lab sessions reinforcing core concepts such as memory management, concurrency, systems programming, and algorithms.

Lead Web Systems Engineer

Feb 2024 – Aug 2025

OwlHacks

Philadelphia, PA

- Led development of Temple University's official hackathon website using TypeScript and Next.js under tight deadline constraints.
- Coordinated a team of 3 developers/designers, managing implementation and feature delivery via Slack.

PROJECTS

Game Boy Color Emulator

- Built a cycle-accurate emulator for the Game Boy Color using C with cross-platform rendering/input via SDL3.
- Implemented the Sharp SM83 CPU core, including full instruction decoding, register management, flag behavior, and precise timing synchronization with hardware cycles.
- Recreated the system's memory architecture, including memory-mapped I/O, bank switching (MBC), and synchronization between CPU, GPU (PPU), and timers.

CHIP-8 Interpreter

- Implemented a CHIP-8 interpreter in C++, including full opcode decoding, register operations, stack management, timers, and input handling.
- Built the graphics and input layer using SDL3, emulating the CHIP-8 display, pixel rendering, and keyboard mapping.

Smart Mirror (Project Lumina)

- Built Raspberry Pi-based system integrating Python backend services with modular frontend interface.
- Designed system architecture for real-time data display (weather, calendar, notifications, etc.) with persistent configuration storage.
- Developed a modular frontend allowing remote configuration via a Progressive Web App (PWA).

SKILLS

Languages / Frameworks: C, C++, Python, ASM (Z80, MIPS), TypeScript/JavaScript, C#, .NET, LaTeX, Vimscript

Development Tools / Platforms: Docker, Make, CMake, Node, Git, Ninja, Vim, Neovim, Visual Studio (Code & Enterprise), GitHub, GitHub Actions